

FraudForge AI

Comprehensive Solution Walkthrough

Closed-Loop Adversarial Payment Defense & Zero-Day Generalization Framework

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Synthetic Defensive Research & Simulation Disclaimer

FraudForge AI is an advanced synthetic defensive simulation and adversarial machine learning research prototype.

- All transaction streams, user profiles, and behavioral telemetry are mathematically synthesized using calibrated statistical distributions (Pareto, log-normal, beta, and empirical diurnal curves).
- Attack archetypes and evolved genomes represent synthetic adversarial testing scenarios constructed to proactively probe detector blind spots.
- Policy decision thresholds, risk weights, reason codes, and simulated cost structures demonstrate algorithmic principles and do not represent Mastercard's proprietary production rules, confidential fraud risk parameters, or live global payment network integrations.
- Metrics such as "Money Protected" and "Generalization Gain" represent evaluations on synthetic benchmark datasets (D_test, D_adversarial, D_unseen) and should not be construed as realized financial savings on live payment volumes.

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1. Executive Summary

Modern payment ecosystems face an asymmetric threat: while financial institutions predominantly train fraud detection models on historical fraud registries, threat actors increasingly utilize Generative AI, autonomous payment agents, and polymorphic evasion techniques to craft zero-day attack vectors. Because legacy detectors over-rely on known heuristic cues (such as sudden device changes, high velocity, or anomalous dollar amounts), sophisticated adversaries bypass detection simply by masking these specific signals.

FraudForge AI resolves this systemic asymmetry through a continuous, closed-loop Red Team / Blue Team AI Defense Lab:

1. **Attack Genome Decomposition:** Deconstructs payment attacks into an orthogonal, 10-dimensional behavioral genome across 28 foundational archetypes.
2. **Adversarial Red-Team Discovery:** Algorithmically evolves novel synthetic fraud candidates via mutation and crossover operators, evaluating them directly against the baseline detector to uncover blind spots before fraudsters exploit them.
3. **Adaptive Blue-Team Hardening:** Automatically mines false negatives, extracts feature dependency imbalances, and retrains the detector on synthetic blind spots.
4. **Empirical Zero-Day Generalization:** Demonstrates that adversarial retraining transfers to completely unseen Generation-2 attack variants, improving detection recall from 23.10% to 93.80% (+70.70 percentage points) with a 91.9% reduction in false negatives on synthetic benchmarks.
5. **Multi-Tier Risk-Aware Decision Engine:** Translates calibrated probabilities into four operational payment actions (ALLOW, MONITOR, STEP_UP_AUTH, BLOCK) with transparent, explainable reason codes, reducing transaction friction while preserving intercept rates.

2. Problem Statement & the Blind-Spot Dilemma

2.1 The Asymmetry of Historical-Data Learning

Supervised machine learning algorithms (e.g., XGBoost, LightGBM, Deep Neural Networks) optimize decision boundaries against distributions observed in historical training sets. When trained on conventional payment fraud data, classifiers naturally identify the most prominent historical discriminators:

- `device_change = 1` (New, unrecognized device)
- `transaction_velocity_1h >> baseline` (Rapid card-draining bursts)
- `transaction_amount >> average_customer_amount` (Sudden high-ticket purchases)
- `IP_risk_score > 0.80` (Flagged datacenter proxies or VPNs)

2.2 The Adversarial Blind Spot

When an adversary utilizes Generative AI to execute credential abuse via trusted-device masking, low-and-slow velocity smoothing, or sub-threshold amount structuring, the historical cues vanish. Because the model placed disproportionate decision weight on `device_change`, its detection probability collapses:

$$P(\text{Fraud} \mid x) \approx \sigma(w_{\text{device}} \cdot x_{\text{device}} + w_{\text{amount}} \cdot x_{\text{amount}} + \dots)$$

When `x_device = 0` (masked via virtual device clones) and `x_amount` matches normal distributions, the detector outputs a low risk score ($P < 0.30$). Consequently, the transaction is categorized as ALLOW, creating an exploitable zero-day blind spot.

Adversary Strategy:	Baseline Detector State:	Outcome:
GenAI Behavioral Mimicry + Trusted-Device Masking + Sub-Threshold Amounts	→ Heavy Over-Reliance on "device_change" Signal (Blind to Subtle Telemetry)	→ FALSE NEGATIVE Baseline Recall: 23.10% (769 / 1,000 Missed)

3. FraudForge AI — The Proposed Solution

FraudForge AI introduces a proactive paradigm: **do not wait for real fraud losses to occur to collect training labels**. Instead, use algorithmic red-teaming to synthesize defensive blind-spot variants, evaluate detection failures in a sandboxed laboratory, and harden the model before deployment.

Core Closed-Loop Pipeline

1. IDENTIFY	Catalog & 10-Dimensional Attack Genome
↓	
2. RED-TEAM GENERATION	Genetic Discovery Engine - Mutation, Crossover & Domain Compatibility Rules - Multi-Objective Scoring (Novelty, Realism, Evasion)
↓	
3. EVALUATION	Baseline XGBoost Classifier Testing - Synthetic Transaction Synthesis via Distributions - Granular Per-Vector Detection & Blind Spot Mining
↓	
4. BLUE-TEAM HARDENING	Retraining on Blind Spots - Augmented Adversarial Dataset Formulation - Feature Dependency Rebalancing
↓	
5. RE-ATTACK BENCHMARK	Unseen Gen-2 Generalization - Dataset D (Fresh Seeds, Evolved Gene Combinations) - +70.70 pt Recall Gain / 91.9% Fewer Misses
↓	
6. OPERATIONALIZATION	Risk Decision Engine - ALLOW / MONITOR / STEP_UP_AUTH / BLOCK - Calibrated Probabilities & Transparent Reasons

4. System Architecture

The FraudForge AI codebase is organized into decoupled layers separating core ML mathematics, synthetic generation, API services, and user interfaces:

FraudForge-AI/	
src/	
attacks/	Red-Team Attack Intelligence
attack_library.py	28 Cataloged Archetypes & Query Interface
attack_genome.py	10-Dimensional Genome Vocabulary & Mappings
novelty_engine.py	Multi-Objective Scoring (Novelty, Realism, Evasion)
attack_discovery.py	Genetic Discovery Engine & Simulator Integration
attack_mutator.py	Targeted Perturbation & Signal Masking Engine
simulation/	Synthetic Payment Simulation
distributions.py	Diurnal curves, Pareto tails & category risk priors
transaction_generator.py	22-Feature Statistical Transaction Generator
features/	Preprocessing & Transformations
feature_engineering.py	RobustScaler, OneHotEncoder & Pipeline Artifacts

— detection/	Blue-Team Machine Learning Detectors
— train.py	Baseline XGBoost Classifier Training
— predict.py	Real-Time Scoring & Inference
— evaluate.py	Granular Per-Attack Recall Analytics
— risk_engine.py	Multi-Tier Policy Engine & Reason Code Generator
— threshold_analysis.py	Cost-Curve Sweeps & ECE Calibration Auditing
— mitigation.py	Automated Payment Mitigation Actions
— audit_logger.py	Structured JSONL / CSV Audit Logging
— adversarial/	Closed-Loop Feedback & Hardening
— feedback_loop.py	3-Dataset Adversarial Hardening Pipeline
— zero_day_hardening.py	Zero-Day Retraining & Gen-2 Benchmark Evaluation
— api/	Production-Style REST API Layer
— app.py	FastAPI Application & CORS Configuration
— schemas.py	Pydantic Schemas for Strict Input/Output Validation
— routes/	Modular Endpoints (catalog, discovery, hardening, risk)
— utils/	
— config.py	Global Constants, Schema Definitions & File Paths
— frontend/	
— index.html	Single-Page Enterprise Security Dashboard
— models/	Serialized ML Artifacts (.joblib)
— experiments/	Verified Empirical Benchmark JSON / CSV Reports
— tests/	72 Unit & Integration Tests (100% Passing)

5. Machine Learning Methodology & Feature Pipeline

5.1 Synthetic Dataset Design (22 Features)

The simulation pipeline synthesizes correlated multi-modal transaction records:

Feature Category	Dim.	Feature Names	Description & Distribution
Financial	3	transaction_amount, average_customer_amount, amount_deviation	Log-normal spending with Pareto heavy tails; relative spending ratios
Temporal	3	transaction_hour, transaction_velocity_1h, transaction_velocity_24h	Diurnal transaction frequencies; sliding- window burst rates
Device & Network	4	account_age_days, device_age_days, device_change, IP_risk_score	Hardware registration lifespan; IP reputation priors
Merchant & Channel	2	merchant_category, payment_channel	Categorical surface mappings (12 merchant types, 7 payment channels)
Identity & Auth	4	identity_risk_score, behavioral_deviation, failed_authentication_count, authentication_method	KYC/synthetic identity indicators; biometric timing cadence anomalies
Geographic	4	geographic_deviation, transaction_country, customer_country, merchant_risk_score	Cross-border location mismatches; corridor risk priors
Ground Truth	2	fraud_label, attack_type	Binary classification target ($y \in \{0,1\}$) and archetype label

5.2 Preprocessing & Robust Transformation

Features undergo deterministic preprocessing via FraudFeaturePipeline (src/features/feature_engineering.py):

- **Numerical Scaling:** RobustScaler scales features using median and interquartile range (IQR), preventing extreme outlier dollar amounts from distorting gradient calculations: $x_scaled = (x - \text{median}(x)) / \text{IQR}(x)$.
- **Categorical Encoding:** OneHotEncoder(handle_unknown='ignore', sparse_output=False) encodes non-ordinal strings (merchant_category, payment_channel, authentication_method, transaction_country, customer_country).
- **Feature Count:** Transformed feature vector expands to 53 numerical dimensions for gradient-boosted trees.

5.3 Detector Architecture

- **Primary Model:** Gradient-Boosted Decision Trees (XGBClassifier, n_estimators=100, max_depth=5, learning_rate=0.10, subsample=0.80, eval_metric='logloss').
- **Probability Estimation:** Calibrated logistic output representing $P(\text{Fraud} = 1 \mid x)$.

6. The 10-Dimensional Attack Genome Matrix

Rather than treating fraud attacks as opaque, monolithic scripts, FraudForge AI deconstructs attack techniques into an orthogonal 10-dimensional genomic space:

#	Gene	Vocabulary
1	target	account, identity, merchant, payment_agent, checkout, payment_authorization, customer
2	entry_vector	credential_abuse, prompt_manipulation, deepfake_synthesis, social_engineering, session_hijacking, api_exploit, malware
3	behavior	low_and_slow, burst_velocity, behavioral_mimicry, cross_channel, cart_manipulation, identity_fabrication
4	evasion_strategy	trusted_device_masking, velocity_smoothing, amount_camouflage, biometric_spoofing, risk_signal_suppression
5	payment_channel	e-commerce, p2p_transfer, mobile_app, api_gateway, pos_chip, pos_contactless, recurring_subscription
6	amount_strategy	normal_looking, micro_transactions, high_ticket_escalation, sub_threshold_structuring, stealth_discounted
7	temporal_strategy	distributed, off_hours, daytime_peak, rapid_burst
8	identity_strategy	existing_account, synthetic_sleeper, fabricated_identity, delegated_agent, swapped_sim, authorized_victim
9	merchant_strategy	single_target, mcc_rotation, high_risk_crypto, digital_goods
10	geographic_strategy	domestic_matching, cross_border_arbitrage, proxy_spoofed

All 28 known archetypes (across AI Social Engineering, Synthetic Identity, Automated ATO, Micro-Transactions, AI Agent Exploits, Cross-Channel Evasion, and Merchant Exploits) are cataloged as ground-truth genome configurations.

7. Red-Team Discovery & Blue-Team Hardening Loop

7.1 Novel Attack Discovery (Red-Team)

The genetic discovery engine (src/attacks/attack_discovery.py) searches the 10-dimensional space:

- 6. **Genetic Mutation & Crossover:** Combines traits from known archetypes while mutating parameters.
- 7. **Domain Compatibility Filter:** Enforces 9 business logic rules (e.g., physical POS contactless cannot execute API gateway prompt injections without cross-channel mechanics).
- 8. **Near-Duplicate Pruning:** Enforces minimum weighted gene distance (>15% novelty vs. known archetypes).
- 9. **Multi-Objective Transparent Scoring:** $Priority = 0.40 \times Novelty + 0.30 \times Realism + 0.30 \times Evasion\ Potential$.
- 10. **Detector Evaluation:** Synthesizes transactions for each candidate and tests them against the baseline model, isolating those with low detection recall as CRITICAL BLIND SPOTS (<50% recall).

7.2 Adaptive Hardening & Gen-2 Generalization (Blue-Team)

- 11. **False-Negative Mining:** Isolates candidates where baseline recall is lowest (e.g., NSA-001, NSA-003, NSA-008, NSA-012, NSA-004).
- 12. **Adversarial Dataset Augmentation:** Synthesizes augmented training datasets ($D_{train_hardened} = D_{train} \cup D_{blind_spots}$).
- 13. **Model Retraining:** Trains the hardened XGBoost classifier on rebalanced feature weights.
- 14. **Generalization Benchmark (Dataset D):** To verify that the model did not simply overfit or memorize specific training patterns, the hardened detector is tested against completely fresh, unseen Generation-2 variants (seed=2026) featuring altered gene structures and different transaction amounts.

8. Verified Empirical Benchmark Results

All metrics below are drawn directly from the verified experiment outputs (experiments/zero_day_hardening_summary.txt, experiments/zero_day_hardening_report.json, and experiments/risk_policy_evaluation.json):

8.1 Zero-Day Hardening: Training Candidates (Gen-1)

Candidate ID	Attack Profile	Baseline	Hardened	Δ
NSA-001	Trusted-Device Omnichannel Social Engineering Trap	10.50%	97.50%	+87.00 pts
NSA-003	Trusted-Device Low-and-Slow Merchant Return Arbitrage	7.00%	100.00%	+93.00 pts
NSA-008	Biometric-Injected Behavioral Mimicry Account Takeover	11.50%	98.00%	+86.50 pts
NSA-012	Camouflaged Cart Manipulation Social Engineering Trap	27.50%	100.00%	+72.50 pts

Candidate ID	Attack Profile	Baseline	Hardened	Δ
NSA-004	Signal-Suppressed Low-and-Slow Checkout Exploit	54.00%	100.00%	+46.00 pts

8.2 Generalization Benchmark on Fresh Unseen Gen-2 Variants (Dataset D)

Variant ID	Gen-2 Attack Profile	Baseline	Hardened	Gen. Gain
NSA-001-V2	Trusted-Device Omnichannel Social Engineering (Gen-2)	14.50%	79.00%	+64.50 pts
NSA-003-V2	Trusted-Device Low-and-Slow Return Arbitrage (Gen-2)	7.00%	100.00%	+93.00 pts
NSA-008-V2	Biometric-Injected Behavioral Mimicry ATO (Gen-2)	37.00%	99.00%	+62.00 pts
NSA-012-V2	Camouflaged Cart Manipulation Social Trap (Gen-2)	1.50%	91.00%	+89.50 pts
NSA-004-V2	Signal-Suppressed Low-and-Slow Checkout (Gen-2)	55.50%	100.00%	+44.50 pts
OVERALL	Overall Unseen Gen-2 Benchmark (1,000 Tested)	23.10%	93.80%	+70.70 pts

Key Takeaway: On 1,000 unseen Gen-2 transactions, missed attacks dropped from 769 down to 62 (-707 misses, a 91.9% reduction in false negatives).

8.3 Defender Regression Check (Normal Legitimate Payments)

Metric	Baseline Detector	Hardened Detector	Delta
Legitimate Transaction Recall	96.33%	98.33%	+2.00%
Legitimate Transaction Accuracy	98.65%	97.30%	-1.35%
False Positive Rate (FPR)	0.94%	2.88%	+1.94%

9. Risk-Aware Decision Engine & Explainability

9.1 Multi-Tier Operational Policy

Rather than forcing a binary 0/1 decision, RiskDecisionEngine maps calibrated probabilities into four actionable tiers:

Probability Range	Risk Tier	Action	Operational Mechanism
[0.00 – 0.30)	LOW	ALLOW	Frictionless payment authorization
[0.30 – 0.60)	MEDIUM	MONITOR	Approved; background telemetry heightened
[0.60 – 0.85)	HIGH	STEP_UP_AUTH	Friction challenge (3DS v2, Biometric Push, OTP)
[0.85 – 1.00]	CRITICAL	BLOCK	Immediate autonomous rejection

9.2 Policy Optimization Comparison (Simulated Relative Cost)

Policy Configuration	Intercept Rate	Block Recall	FPR	Step-Up	Cost Index
Binary Baseline ($P \geq 0.50$)	99.30%	99.30%	3.31%	0	38.0
BALANCED Policy (Multi-Tier)	99.30%	80.00%	1.06%	19	23.8 (-37.4%)
STRICT_SECURITY Policy	99.30%	85.00%	1.54%	22	28.5 (-25.0%)

Cost model: $Cost = 10.0 \times FN + 1.0 \times FP + 0.25 \times Step-Up$. By challenging rather than declining ambiguous transactions, the Balanced policy cuts false declines by over 65% while maintaining a 99.30% intercept rate.

9.3 Transparent Reason Codes

Every decision provides human-interpretable, feature-grounded reason codes:

- **HIGH_IP_RISK:** $IP_risk_score \geq 0.70$
- **UNUSUAL_AMOUNT:** $amount_deviation \geq 2.0$
- **HIGH_VELOCITY:** $velocity_1h \geq 5$
- **DEVICE_CHANGE:** $device_change = 1$
- **BEHAVIORAL_DEVIATION:** $behavioral_deviation \geq 0.60$
- **IDENTITY_RISK:** $identity_risk_score \geq 0.65$

10. REST API Architecture

The Python backend exposes a modular, documented FastAPI service running on `http://127.0.0.1:8000`:

Endpoint	Method	Purpose	Key Input / Output
/api/v1/health	GET	Service & model artifact health check	Status, loaded models, archetype count
/api/v1/catalog/known-attacks	GET	Returns 28 cataloged archetypes	Full catalog list with genomes & severity
/api/v1/catalog/genome-vocabulary	GET	Returns all 10 genome gene values	Dictionary of available genome choices
/api/v1/discovery/generate-candidate	POST	Evolves synthetic attack candidate	Custom/mutated genome & multi-objective scores
/api/v1/discovery/evaluate-candidate	POST	Tests candidate against baseline detector	Real detection recall, missed count, blind spot tier
/api/v1/hardening/compare-defense	POST	Compares baseline vs. hardened detector	Baseline vs. hardened recall & generalization gain
/api/v1/risk/evaluate-transaction	POST	Scores a single payment transaction	Probability, risk score, tier, action, reason codes
/api/v1/risk/evaluate-batch	POST	Batch transaction scoring	Array of decisions & summary telemetry

Interactive OpenAPI documentation is automatically available at <http://127.0.0.1:8000/docs>.

11. Interactive Frontend Architecture

The frontend (`frontend/index.html`) is a responsive, single-page application served on `http://localhost:3000`:

- **Command Center:** Real-time transaction stream, threat telemetry counters, radar charts, and live security audit logs.
- **Attack Lab:** Interactive 10-dimensional genome builder, mutation intensity slider, and live detector testing.
- **Defense Lab:** Side-by-side comparison of baseline detector vs. hardened detector, 9-stage adaptive pipeline animation, and empirical generalization metrics.
- **Attack Genome Matrix:** Full interactive matrix visualizer for all 28 cataloged archetypes.
- **Risk Decision Engine:** Live single-transaction scoring sandbox, multi-tier policy threshold sliders, and reason code breakdown.
- **90-Second Judge Demo:** Automated step-by-step walkthrough presenting the complete core narrative in under 90 seconds.

12. Verification & Automated Test Suite

FraudForge AI includes a comprehensive 72-test test suite executed via `pytest`:

```
===== 72 passed in 16.28s =====
```

- **API Tests (`test_api.py`):** Validates health, catalog retrieval, candidate generation, candidate evaluation, defense comparison, and risk scoring.
- **Discovery Tests (`test_attack_discovery.py`, `test_attack_genome.py`, `test_novelty_engine.py`):** Validates genome roundtrip serialization, deterministic genetic operators, scoring bounds, and distance metrics.
- **Simulation Tests (`test_transaction_generator.py`):** Validates schema invariance, absence of nulls/NaNs, range constraints, and realistic feature divergence.
- **Detection & Policy Tests:** (`test_detection.py`, `test_feature_engineering.py`, `test_risk_engine.py`, `test_threshold_analysis.py`) Validates XGBoost training/inference, probability calibration, risk tier ordering, and reason code generation.
- **Adversarial Loop Tests:** (`test_feedback_loop.py`, `test_zero_day_hardening.py`, `test_benchmarks.py`) Validates 3-dataset closed-loop retraining, Gen-2 creation, and target leakage prevention.

13. Limitations & Future Work

Limitations

15. **Synthetic Data Boundaries:** While generator distributions have been calibrated for realistic variance and heavy-tail overlap, real-world cardholder behavior exhibits complex temporal and macroeconomic shifts.
16. **Tabular Feature Representation:** The current architecture evaluates structured tabular telemetry. Future implementations could incorporate graph neural networks (GNNs) for multi-entity mule network analysis.
17. **Static Adversarial Feedback:** Red-team exploration is driven by genetic algorithms and heuristics rather than deep reinforcement learning agents.

Future Roadmap

18. **Multi-Agent Reinforcement Learning (MARL):** Deploy competitive LLM/RL agents playing a continuous minimax game over payment authorization graph embeddings.
19. **Graph-Level Payment Swarm Detection:** Extend the 10-dimensional genome to model multi-hop account topology and distributed money mule networks.
20. **Federated Adversarial Learning:** Enable privacy-preserving cross-bank collaboration where synthetic blind-spot vectors are shared without exposing raw customer data.

14. Conclusion

FraudForge AI provides a concrete, empirical answer to the challenge of Generative AI payment fraud. By structuring attacks into a 10-dimensional genome, discovering detector blind spots algorithmically, and executing adaptive retraining, FraudForge AI transforms fraud prevention from a reactive post-loss investigation into a proactive, self-hardening defense system.

Verified at 93.80% recall (+70.70 pt gain) on completely unseen Generation-2 attacks, FraudForge AI proves that machine learning models can anticipate and neutralize adversarial payment techniques before they reach production payment networks.